

Demonstration of Higher-Order Tensor Contraction via Optical Nonlinear Wave Mixing

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Abstract We demonstrate direct optical higher-order tensor contraction using nonlinear wave mixing in a periodically poled lithium niobate waveguide. The system performs the contraction of three order-3 tensors over three pairs of indices at 5 GHz with ~4-bit precision. ©2026 The Author(s)

Introduction

The growth of artificial intelligence and high-dimensional data processing has driven the need for hardware that is capable of handling multidimensional data structures. These structures can be represented by tensors, which generalize vectors and matrices to higher dimensions (e.g., an order-1 tensor is a vector, and an order-2 tensor is a matrix) [1]. As shown in Fig. 1(a), an order- M tensor requires M indices to specify an element (e.g., A_i , A_{ij} , and A_{ijk} for orders-1, 2, and 3, respectively). One fundamental tensor operation is tensor contraction, illustrated in Fig. 1(b), which is the higher-order analogue of matrix multiplication [2]. This operation sums over shared indices across multiple tensors to produce a lower-order result. Tensor contraction is used in many advanced applications, including machine learning, quantum computing and simulations, and hyperspectral imaging [3-5].

Tensor contractions are typically implemented using sequences of matrix operations on conventional electronic hardware. Such approaches rely

on reshaping or decomposing tensors into lower-order representations, which significantly increases memory overhead and computational complexity [2]. To the best of our knowledge, there has been no demonstration of directly performing higher-order tensor contraction without decomposition. Optical computing can potentially overcome this limitation by leveraging multiple degrees of freedom of light in order to simultaneously operate over multiple tensor indices [6].

Previous work in optical computing has primarily focused on matrix-vector (MVM) and matrix-matrix multiplication (MMM), implemented using architectures with linear optical responses [7-13]. While these systems can perform MVM and MMM, extending them to higher-order tensor contraction might be challenging. Nonlinear optical effects based on second- and third-order susceptibilities provide a potential pathway, as they enable physical interactions among multiple input signals simultaneously. While MVM and MMM based on nonlinear interactions have been demonstrated [14-17], direct implementation of higher-order tensor contraction has not yet been shown. A laudable goal, therefore, might be to perform tensor contraction directly on native tensor representations without decomposition by using nonlinear optical effects.

In this work, we demonstrate optical higher-order tensor contraction using nonlinear wave mixing. We map the indices of three order-3 tensors (A , B , and C) to the frequency and time domains. We perform the contraction of the input tensors over three pairs of indices simultaneously in a periodically poled lithium niobate (PPLN) waveguide. Our approach utilizes cascaded sum-frequency generation (SFG) and difference-frequency generation (DFG) to coherently compute the summation terms for the output element in a single pass through the PPLN. We experimentally validate this approach by contracting three order-3 tensors of size $2 \times 2 \times 2$ at 5 GHz, achieving ~4-bit precision and providing a path toward higher-order photonic computing.

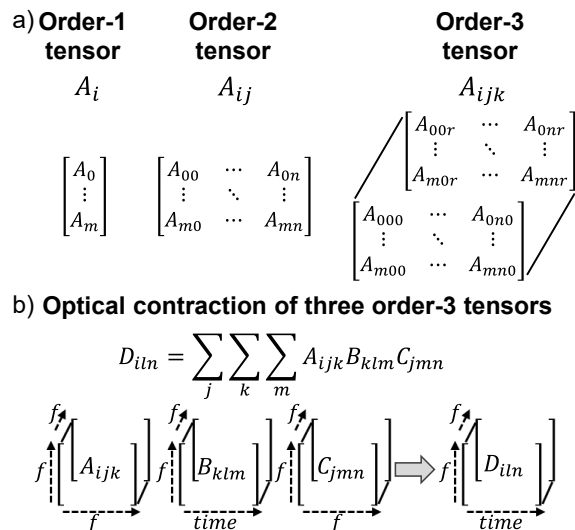


Fig. 1: (a) Tensors of order-1, order-2, and order-3. (b) Contraction of three order-3 tensors by mapping the elements to the frequency and time domain to compute optically.

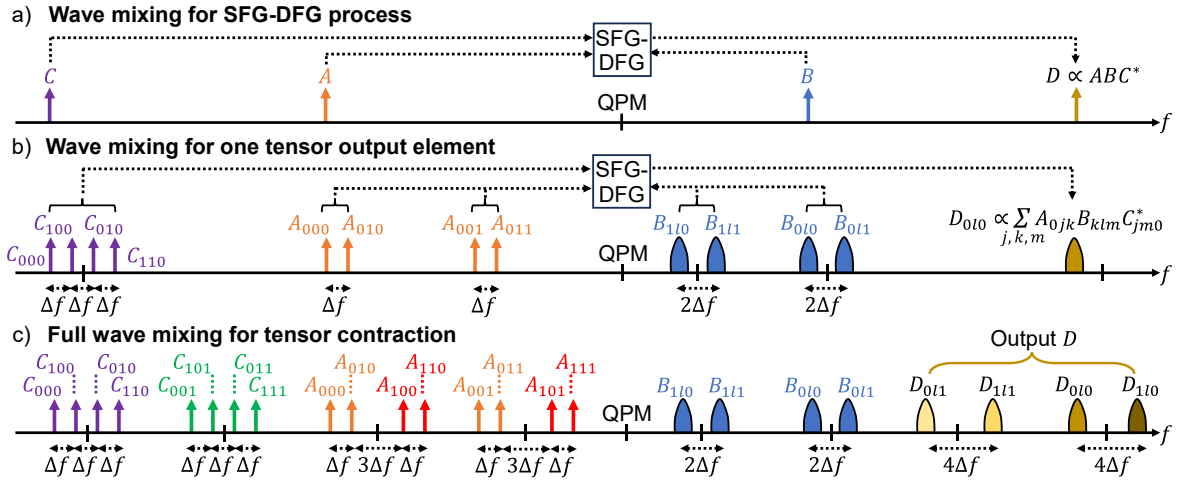


Fig. 2: Concept of all-optical tensor contraction by using nonlinear wave mixing. (a) Wave mixing diagram for SFG-DFG process. (b) Wave mixing diagram for one output tensor element. (c) Wave mixing diagram the full tensor contraction, each tensor with a size of $2 \times 2 \times 2$. SFG-DFG: sum-difference frequency generation. QPM: quasi-phase matching.

Concept

Figure 2 illustrates the concept of optical tensor contraction using nonlinear wave mixing. In our approach, each tensor element is encoded as the amplitude and phase of the input optical waves. For an order-3 tensor of size $N \times N \times N$, such as A , there are three indices (i, j, k) to describe the elements of the tensor in each dimension. Here, we set the input tensors size to $N = 2$, where each tensor will have 8 elements. In this work, we contract the three input tensors A , B , and C over three pairs of indices to obtain the output D , which is an order-3 tensor where its elements are $D_{ilm} = \sum_j \sum_k \sum_m A_{ijk} B_{klm} C_{jmn}$ (Fig. 1(b)). Each output tensor element will therefore be a summation of N^3 terms. For example, the output $D_{000} = A_{000}B_{000}C_{000} + A_{000}B_{001}C_{010} + A_{010}B_{000}C_{100} + A_{010}B_{001}C_{110} + A_{001}B_{100}C_{000} + A_{001}B_{101}C_{010} + A_{011}B_{100}C_{100} + A_{011}B_{101}C_{110}$.

For the optical implementation of this tensor contraction, the index l is mapped to the time domain at the symbol rate, while the other indices are mapped to the frequency domain (Fig. 1(b)). In this work, we use nonlinear wave mixing in a PPLN waveguide to obtain the output tensor D via multiple simultaneous cascaded SFG and DFG processes [18]. Each SFG-DFG process can be described as $E_D \propto E_A E_B E_C^*$ where the output is located at $f_D = f_A + f_B - f_C$ (Fig. 2(a)). Figure 2(b) shows the eight SFG-DFG processes that occur simultaneously to obtain one output tensor element D_{010} . By selecting the frequency spacing between the input terms as shown in the figure, the eight summation terms of one output tensor element will coherently add at the same output frequency. To obtain the other output elements of D ($D_{011}, D_{110}, D_{111}$) at different frequencies, we encode the remaining input terms at the frequencies shown in Fig. 2(c). As mentioned

above, the index l is serialized in the time domain. Therefore, the first four elements of D ($D_{000}, D_{001}, D_{100}, D_{101}$) are obtained at one time step, and the next four elements of D ($D_{010}, D_{011}, D_{110}, D_{111}$) are obtained at the next time step.

Experimental Setup

Figure 3(a) shows the experimental setup. A mode-lock laser frequency comb is used to generate the carriers for the input tensors A , B , and C . A liquid-crystal-on-silicon (LCoS) programmable filter is used to select the lines which are used as carriers for the input tensors. The tensors A

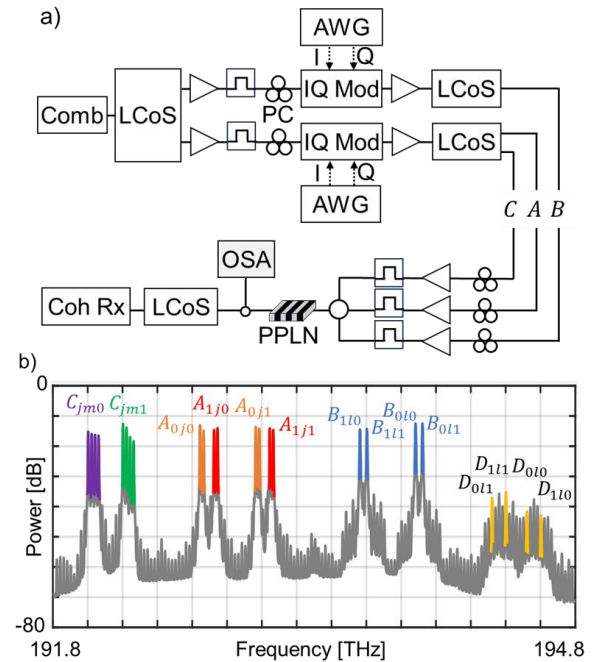


Fig. 3: (a) Experimental setup for optical tensor contraction. (b) Optical spectrum at the output of the PPLN waveguide. AWG: arbitrary waveform generator. IQ Mod: in-phase and quadrature modulator. LCoS: liquid-crystal-on-silicon. PC: polarization controller. OSA: optical spectrum analyzer. Coh. Rx: coherent receiver.

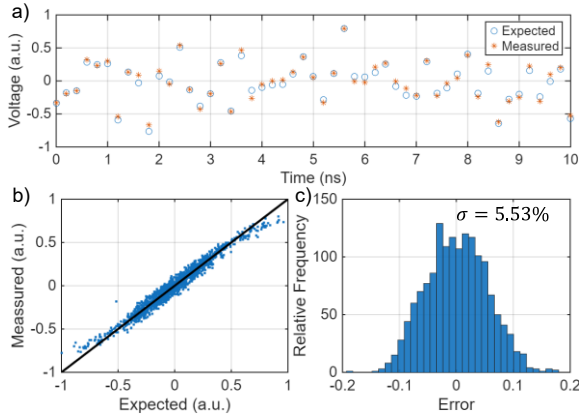


Fig. 4: Experimental results for optical tensor contraction of output D_{011} at 5 GHz. (a) Expected vs measured symbols over 10 ns. (b) Correlation plot of expected vs measured results over 1536 symbols. (c) Histogram of errors over 1536 symbols, with standard deviation of 5.53%.

and C are generated by an arbitrary waveform generator (AWG) that drives an in-phase and quadrature (IQ) modulator, which produces multiple coherent sinusoidal tones around each carrier at Δf (20 GHz) spacing, where each frequency tone carries one element of the tensor. The LCoS selects the lines and adjusts the amplitude and phase of each frequency, allowing for the encoding of different values on the input tensor elements. The tensor B is generated by another AWG that drives another IQ modulator, which produces the time multiplexed data (5 GHz) with a spacing of $2\Delta f$ around each carrier. The inputs are amplified and coupled into the PPLN waveguide. The total input power to the PPLN is 22 dBm. The output conversion efficiency is ~ 20 dB. The PPLN output spectrum is shown in Fig. 3(b). A LCoS at the PPLN output selects the frequency corresponding to each tensor output element. The outputs are captured with a coherent receiver to recover their amplitude and phase values.

Experimental Results and Discussion

Figure 4 shows the experimental results of our optical tensor contraction approach at 5 GHz. In this work, we contract the input tensors A , B , and C (all of size $2 \times 2 \times 2$), to obtain the output D (size of $2 \times 2 \times 2$). At each time sample, the l index of B changes and we obtain 4 elements of D simultaneously (D_{010} , D_{001} , D_{110} , D_{111}). The input tensor B is a sequence of 768 randomly generated $2 \times 2 \times 2$ tensors where the elements are analog values ranging from -1 to 1, while all the tensor elements of A and C are set to 1. In Fig. 4, we show one output of our optical tensor contraction, D_{011} , at 5 GHz. The expected and measured waveforms are plotted for 10 ns in Fig. 4(a). The correlation plot between expected vs measured results are shown across 1536 samples in Fig. 4(b). The histograms of the errors are also shown in Fig. 4(c), where the measured values show good

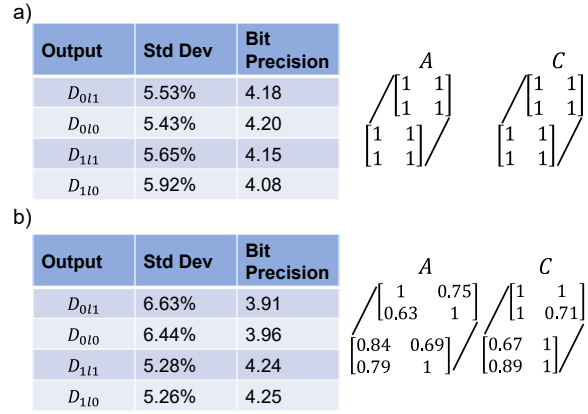


Fig. 5: (a) Standard deviation and bit precision for the four outputs of D when the elements of A and C are all ones. (b) Standard deviation and bit precision for the four outputs of D when the elements of A and C have different values, as shown in the figure.

agreement with the expected values. In this work, the system performs 8 multiply-accumulate (MAC) operations per output element, yielding an effective throughput of 160 GMAC/s at 5 GHz.

We demonstrate the reconfigurability of our system in Fig. 5, where we change the weights of A and C . In Fig. 5(a), we show all the outputs of D when the elements of A and C are all ones. The standard deviation of errors between the measured and the expected waveforms are shown. The standard deviation ranges from 5.92% to 5.43%. This corresponds to a bit precision of 4.08- to 4.20-bits [19]. In Fig. 5(b), the elements of tensors A and C are changed according to the values in the figure by adjusting the amplitude and phase of each frequency with the LCoS. The standard deviation ranges from 6.63% to 5.26%. This corresponds to a bit precision of 3.91- to 4.25-bits. The difference in error may be caused by the different nonlinear conversion efficiency of the PPLN at different input power levels. In our system, scaling to larger N is primarily limited by the phase-matching bandwidth of the PPLN and the available frequency grid.

In summary, we have experimentally demonstrated optical higher-order tensor contraction using nonlinear wave mixing in a PPLN waveguide. By mapping tensor indices onto frequency and time domains and leveraging cascaded SFG-DFG processes, we coherently generated the multiply-accumulate output terms of an order-3 tensor contraction in a single nonlinear waveguide. We experimentally validated the contraction of three $2 \times 2 \times 2$ tensors at 5 GHz, achieving approximately 4-bit precision and yielding a throughput of 160 GMAC/s. This work establishes nonlinear photonics as a viable platform for directly implementing higher-order tensor operations and provides a pathway toward scalable, high-dimensional optical computing architectures.

Acknowledgements

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